```
import React, { useState } from "react";

export default function ColorPaletteGenerator() {
  const [color, setColor] = useState("#1a07ecff");

  // Generate random hex color
  const generateColor = () => {
    const randomColor = "#" + Math.floor(Math.random() * 16777215).toString(16).padStart(
    setColor(randomColor);
  };

  // Copy to clipboard
  const copyToClipboard = () => {
    navigator.clipboard.writeText(color);
    alert("Copied: " + color);
  };

  return (
    <div className="h-screen flex flex-col items-center justify-center" style={{ backgrou
      <h1 className="text-2xl font-bold mb-4">Color Palette Generator</h1>

      <button
        onClick={generateColor}
        className="px-6 py-2 bg-black text-white rounded-lg mb-4"
      >
        Generate Color
      </button>
```

colorpal.jsx > ColorPaletteGenerator

```
ıte } from "react";

ɔn ColorPaletteGenerator() {
.or] = useState("#1a07ecff");

ıex color
 = () => {
 = "#" + Math.floor(Math.random() * 16777215).toString(16).padStart(6, '0');
.or);



d = () => {
d.writeText(color);
- color);



-screen flex flex-col items-center justify-center" style={{ backgroundColor: color }}>
:ext-2xl font-bold mb-4">Color Palette Generator</h1>


ateColor}
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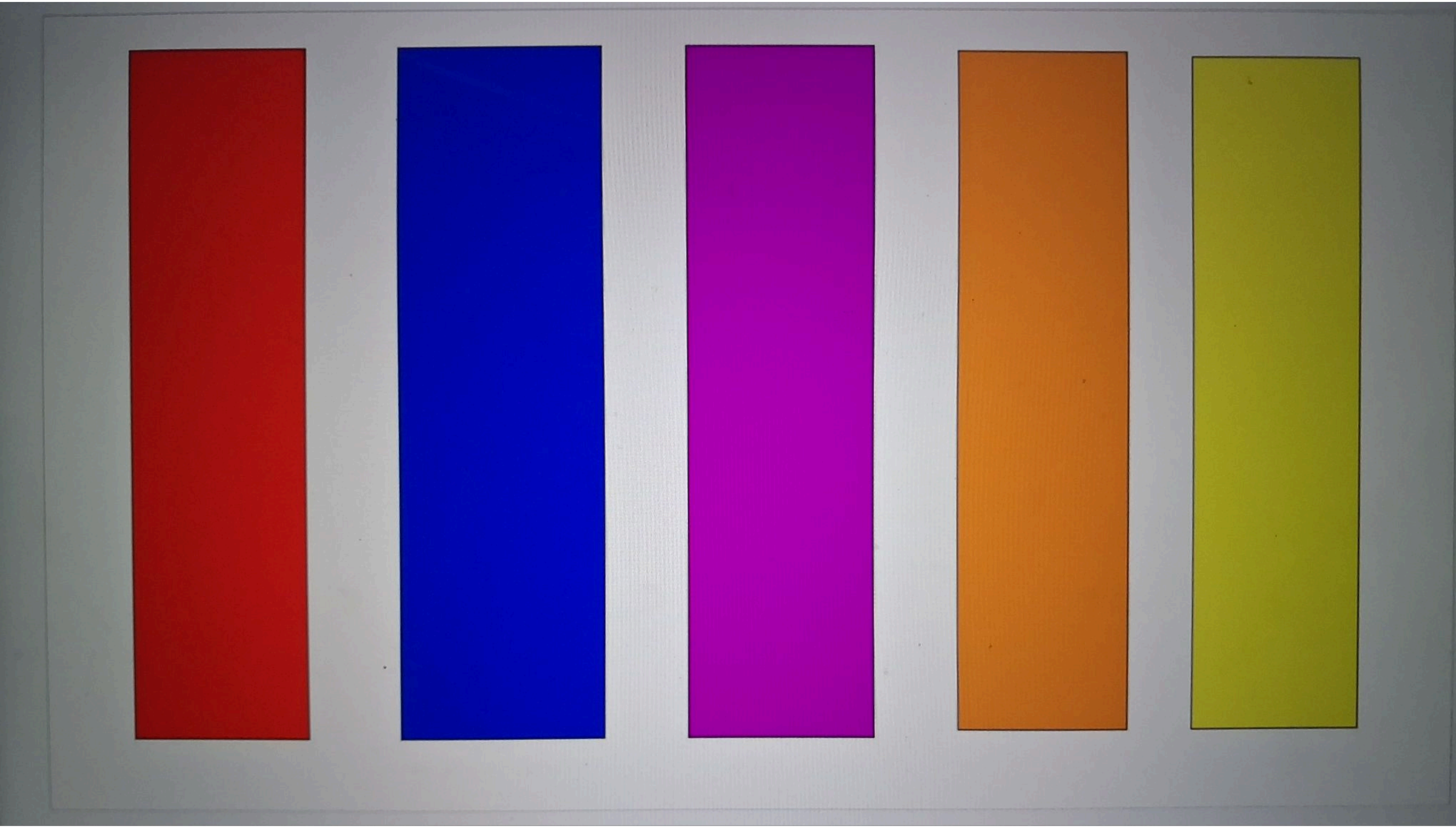
ColorPaletteGenerator

```jsx
export default function ColorPaletteGenerator() {

  return (
    <div className="h-screen flex flex-col items-center justify-center" style={{ backgrou
      <h1 className="text-2xl font-bold mb-4">Color Palette Generator</h1>

      <button
        onClick={generateColor}
        className="px-6 py-2 bg-black text-white rounded-lg mb-4"
      >
        Generate Color
      </button>

      <div
        onClick={copyToClipboard}
        className="cursor-pointer text-xl font-mono bg-white px-4 py-2 rounded-lg shadow"
      >
        {color}
      </div>

      <p className="mt-2 text-sm">Click the color code to copy</p>
    </div>
  );
}
```

```
shivam@Shivams-MacBook-Air ~ % source /Users/shivam/.venv/bin/activate
(.venv) shivam@Shivams-MacBook-Air ~ % cd Desktop/react/my-project
(.venv) shivam@Shivams-MacBook-Air my-project % npm i

up to date, audited 153 packages in 812ms

36 packages are looking for funding
  run `npm fund` for details

2 vulnerabilities (1 moderate, 1 high)

To address all issues, run:
  npm audit fix

Run `npm audit` for details.
(.venv) shivam@Shivams-MacBook-Air my-project % npm run dev

> my-project@0.0.0 dev
> vite
```

```
2 vulnerabilities (1 moderate, 1 high)

To address all issues, run:
  npm audit fix

Run `npm audit` for details.
(.venv) shivam@Shivams-MacBook-Air my-project % npm run dev

> my-project@0.0.0 dev
> vite


  VITE v8.0.3  ready in 528 ms

  ➜  Local:   http://localhost:5173/
  ➜  Network: use --host to expose
  ➜  press h + enter to show help
```



ITM SKILLS UNIVERSITY

Name - Shivam Sah

Project Name - Colour Pallete in (React JS)

Roll no - 150096725052

Index

1. Problem Statement
2. Your solution
3. Working (Method/Mind map)
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1. Problem Statement

In the field of web design and front-end development, choosing appropriate colors is a critical task that directly impacts the visual appeal, readability, and user experience of an application or website. Designers and developers often spend a significant amount of time experimenting with different color combinations to achieve the desired look and feel. Manually selecting colors and searching for suitable hex codes can be inefficient, repetitive, and sometimes confusing, especially for beginners who are not familiar with color theory or design tools.

There is a need for a simple, fast, and interactive solution that can assist users in generating random colors instantly without requiring advanced design knowledge. Such a tool should not only generate colors but also present them in a user-friendly manner, allowing users to easily view and use the generated values in their projects.

The problem, therefore, is to design and develop a **Color Palette Generator web application using React JS** that can dynamically create random color values and display them in real time. The application should allow users to generate new colors with a single click, view the corresponding hex color codes, and copy these codes directly to the clipboard for immediate use. Additionally, the application should provide a responsive and visually appealing interface that enhances usability and engagement.

This project also aims to demonstrate the practical implementation of core React concepts such as **state management, component re-rendering, and event handling**, while integrating browser functionalities like the Clipboard API. The solution should be efficient, lightweight, and accessible, making it useful for both beginners and experienced developers.

In conclusion, the goal is to build a tool that simplifies the color selection process, improves productivity, and showcases how modern web technologies like React JS can be used to create interactive and user-centric applications.

2. Your solution

Solution: Color Palette Generator (React JS)

How it works

- Generates a random hex color
- Changes the background color
- Displays the hex code
- Copies the code using `navigator.clipboard`

3. Working (Method/Mind map)

1.The application loads with a default background color.
2.When the user clicks the **"Generate Color"** button, a function is triggered.
3.The function generates a **random hex color code** using JavaScript.
4.The generated color is stored in the React state using `useState`.
5.React automatically updates the UI, changing the background color and displaying the new hex code.
6.When the user clicks on the displayed color code, it is copied to the clipboard using `navigator.clipboard.writeText()`.
7.A confirmation message is shown to indicate the color code has been copied.

User Click
↓
Generate Color Function
↓
Random Hex Code Created
↓
Update React State (useState)
↓
UI Re-render
↓
Background Color Changes + Code Displayed
↓
User Clicks Code
↓
Copy to Clipboard
↓
Confirmation Message

4. Conclusion.

The Color Palette Generator developed using React JS is a simple yet highly useful application that addresses the common challenge of selecting suitable colors for web design and development. By automating the process of generating random hex color codes, the application saves time and effort while also encouraging creativity and experimentation with different color combinations.

This project effectively demonstrates the core concepts of React, including **component-based architecture, state management using hooks (`useState`), and event handling**. The dynamic updating of the user interface without page reload highlights the power and efficiency of React in building modern web applications. Additionally, the integration of browser features like the **Clipboard API** enhances user convenience by allowing instant copying of color codes.

The application is designed with simplicity and usability in mind, ensuring that even beginners can easily interact with it. At the same time, it provides a strong foundation for further enhancements such as generating multiple color palettes, adding theme options, saving favorite colors, or improving the UI with advanced styling frameworks.

Overall, this project not only fulfills its primary objective of creating a functional color generator but also serves as a practical learning experience in front-end development. It showcases how basic programming logic combined with React can be used to build interactive, responsive, and user-friendly applications.

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Thank You